REQUEST FOR INFORMATION (RFI) / SOURCES SOUGHT ANNOUNCEMENT

DEPARTMENT OF THE NAVY
Program Acquisition Executive Mission Systems (PAE MS)
Capability Portfolio Executive Combat Systems & Fires (CPE CS&F)
Integrated Combat System (ICS), PMW 630

SECTION 1: GENERAL INFORMATION

Document Type: Request for Information (RFI) / Sources Sought Notice

Solicitation / Reference Number: N6175027R1201

Contracting Activity: Navy PAE Mission Systems, Washington Navy Yard, DC

Program Title: Surface Ship ICS Engineering and Integration

North American Industry Classification System (NAICS) Code: 541330 – Engineering Services (Military and Aerospace Equipment and Military Weapons)

Product Service Code (PSC): AC13 – R&D: Defense Systems: Weapons / R425 – Support - Professional: Engineering/Technical

Response Due Date: 04 December 2026, 15:00 EST

Classification Level: Primary document UNCLASSIFIED. Classified responses or annexes up to SECRET / NOFORN handled by separate coordination (see Section 7).

SECTION 2: STATUTORY DISCLAIMER & SPECIAL NOTICE

This RFI is issued solely for **information and planning purposes** in accordance with **FAR 15.101(c)**.

This notice does **not** constitute a Request for Proposal (RFP), Invitation for Bid (IFB), or an announcement of a forthcoming solicitation. Issuance of this notice does not obligate the Government to issue a formal solicitation or award a contract.

The Government will **not** pay or reimburse any costs incurred by respondents in preparing, submitting, or demonstrating information in response to this notice. All submissions become Government property and will not be returned.

Participation in this RFI is voluntary. Failure to respond will not preclude a vendor from participating in any future competitive acquisition, should one be released.

Respondents may submit capability statements addressing **any subset or all** of the focus areas detailed herein.

SECTION 3: BACKGROUND & PROGRAM CONTEXT

3.1 Background & Context

Modern surface warfare requires rapid, modular capability delivery across the Navy's surface fleet. The Navy's ICS program is making this vision a reality through use of virtualization, Kubernetes orchestration, and development of a common suite of mission applications enabling a "build once and deploy across the fleet" methodology for delivery of capability to frigates, destroyers, amphibious ships, aircraft carriers, future manned and unmanned vessels, and future foreign military sales vessels.

The ICS program is conducting market research to identify industry partners capable of executing comprehensive systems engineering and software integration for U.S. Navy surface combat systems. This scope is currently being executed under the N00024-23-C-5117 contract awarded to Lockheed Martin Rotary & Mission Systems in September 2023.

3.2 Technical Scope & Baselines

The technical scope spans current and next-generation baseline configurations running incremental deliveries of the Integrated Combat System (ICS) software stack (infrastructure services and tactical platform services) and common mission applications. Applicable ship deliveries include the following:

- ICS-Enabled AEGIS Combat System Baselines 9.C3+ and Baseline 10.1+ running on Mk6 Mod0/1 hardware with IaaS ORDALT or on Mk6 ModX hardware
- ICS-Enabled Ship Self-Defense System (SSDS) platforms
- FF(X) 3 and follow
- DDG 1000 class in line with delivery of the Zumwalt Enterprise Upgrade Solution (ZEUS)

- Future USN and FMS platforms

SECTION 4: TECHNICAL REQUIREMENTS & CAPABILITY AREAS SOUGHT

The Navy seeks detailed technical capabilities, corporate competencies, and innovative methodologies across eleven (11) functional areas. Vendors may respond to **any or all** of the following scope areas:

Focus Area 1: Programmatic Engineering, Configuration Management & Governance

- **Configuration Management (CM):** Formulating, baselining, and executing Configuration Management Plans (CMP), software build/release planning, Configuration Control Boards (CCBs), and tracking and adjudicating Software Change Requests (SWCRs).
- **Programmatic & Systems Planning:** Developing, maintaining, and synchronizing project-level Roadmaps, Integrated Master Schedules (IMS), and capability delivery cadences.

Focus Area 2: Systems Engineering & Requirements Architecture

- **Technical Baseline Artifacts:** Developing, updating, and managing formal engineering artifacts:
  - Systems Engineering Plans (SEP).
  - Software Development Plans (SDP) utilizing modern Agile/DevSecOps frameworks.
  - System / Subsystem Definition Documents (SSDD).
  - System / Subsystem Specifications (SSS).
  - Software Product Specifications (SPS).
  - Interface Requirements Specifications (IRS).
  - Requirements Traceability Verification Matrices (RTVM) maintaining end-to-end bidirectional requirement-to-test traceability.
  - System Safety Plans (SSP) conforming to MIL-STD-882E and naval weapon safety requirements.
- **Requirements Allocation:** Decomposing ICS, AEGIS, SSDS, and future combat system Top Level Requirements (TLR) into modular capability building blocks and subsystem specifications across surface ship classes (DDG 51, CG 47, FF(X), DDG 1000, L-Class, CVN, unmanned surface vessels, FMS variants, future platforms).

- **Technical Leadership:** Serving as lead technical authorities in Integrated Product Teams (IPTs), Working Groups (WGs), formal milestone reviews (SRR, PDR, CDR), and leading Technical Interchange Meetings (TIMs) to define and allocate interface requirements.
- **Force-Level Mission Thread Analysis:** Evaluating complex, distributed kill-chain mission threads, establishing functional allocations, and resolving tactical latency challenges.
- **ICS Re-Architecture & Model-Based Systems Engineering (MBSE):** Authoring SysML/UML MBSE models and leading modular product development across the ICS Domain Architecture.
- Sensor Management.
- Vehicle Management.
- Information Management.
- Training Management.
- Tactical Decision Management.
- External Communications Management.
- Weapon Management.
- Display Management.
- Infrastructure Services.

Focus Area 3: Fleet Data Analysis, Feasibility Studies & Gap Assessment

- Conducting engineering feasibility studies, system performance assessments, and trade studies for capability insertion across diverse ship classes and configurations.
- Extracting, parsing, and analyzing live fleet exercise data, weapon telemetry, and real-world tactical events to identify performance anomalies and drive capability enhancement.
- Assessing combat system shortfalls, bottlenecks, and operational gaps, and designing/evaluating candidate architectural and software modifications to close these gaps.

Focus Area 4: Tactical Software Design, Coding, Testing, and Integration

- **Combat System Design activities including analysis and algorithm development**
- **Mission software development, unit testing, and component integration.**
- **Cloud-based and Hardware-in-the-Loop (HWIL)-based testing to include element integration, subsystem testing, and system testing**

- **System Integration:** Engineering and integrating software interfaces for specific sensor, weapon, and C2 programs and capabilities, including but not limited to:
  - Patriot Advanced Capability-3 Missile Segment Enhancement (PAC-3 MSE)
  - The Standard Missile (SM) Family of weapons (including all Air Defense, Ballistic Missile Defense, and Extended Range Active Missile variants)
  - MK-41 VLS-compatible canister-launched weapons (including Evolved SeaSparrow Missile (ESSM), Tomahawk Land Attack Missile (TLAM), and Vertical Launch ASROC (VLA)
  - Rolling Airframe Missile (RAM)
  - Electronic Warfare (EW) systems
  - Phased array and rotating radar systems
  - Evolving to common track management solution across AEGIS and SSDS configurations
  - Containerized payloads
- **Tactical Virtualization Configurations:** Designing, provisioning, benchmarking, and troubleshooting virtualized tactical hardware/software configurations and hypervisor-to-bare-metal runtime environments.

Focus Area 5: Hardware/Software Tech Stack, Microservices & System-of-Systems Integration

- **Third-Party Software & Microservices:** Evaluating commercial and third-party software products, conducting compliance refactoring, and defining standardized, ICS-compliant APIs to support microservices orchestrated via Kubernetes.
- **Hypervisor Integration:** Integrating and tuning alternate hypervisors (e.g., VMware ESXi, KVM, bare-metal container engines) to meet strict real-time deterministic performance requirements.
- **Full-Stack System-of-Systems (SoS):** Ensuring the integrated tech stack (hardware compute blades, switching fabric, hypervisors, orchestrators, microservices, and tactical applications) fulfills holistic combat system operational requirements. Leading technical forums to establish cross-contractor deliverable interfaces, lead-ship integration agreements, and deployment schedules.

Focus Area 6: Modeling & Simulation (M&S)

- Developing, maintaining, validating, and updating high-fidelity simulations and functional simulators.

- Conducting system-level, ship-level, and force-level performance simulations, radar cross-section/propagation trade studies, and engagement dynamics modeling to validate tactical algorithms.

Focus Area 7: Cybersecurity Engineering, IA & Program Protection

- **Cyber Systems Engineering:** Refactoring heritage combat system code to contemporary cybersecurity baselines, integrating modern defensive cyber tools, and ensuring zero loss of tactical capability during cyber retrofits.
- **Information Assurance (IA) Compliance:** Enforcing DoD, DoN, and NAVSEA security policies; documenting DoD Instruction (DoDI) 8510.01 Risk Management Framework (RMF) security controls.
- **Model-Based ATO & Cross Domain Solutions:** Supporting technical reviews and data calls for Cross Domain Solutions (CDS); driving the transition to model-based RMF Authorization to Operate (ATO) artifact generation.
- **Vulnerability Scanning & Red Team Support:** Executing continuous vulnerability management (ACAS, SCAP), static/dynamic code analysis (e.g., SonarQube, Fortify), supporting Red Team assessments, and implementing automated STIG remediation workflows.
- **Program Protection & Fleet Readiness:** Authoring Program Protection Plans (PPP); identifying fleet cyber operational and response training shortfalls and recommending tooling enhancements; leading Cyber TIMs with Participating Acquisition Resource Managers (PARMs).

Focus Area 8: Verification, Validation & Automated Testing

- **Automated Test Architecture:** Authoring test strategies, detailed verification plans, and scripts emphasizing automated testing across unit, component, capability, and system levels.
- **Multi-Tier Testing Execution:** Executing and supporting testing to include unit/component testing, integration testing, multi-element integration testing, regression testing, stability testing, functional, system test and evaluation, and fleet acceptance testing.
- **Certification & Maintainability:** Supporting government-led maintainability evaluations. Defining, capturing, and verifying Objective Quality Evidence (OQE) to support formal combat system certification.
- **Defect Adjudication:** Verifying software problem resolutions and supporting hardware failure reporting, analysis, and corrective action systems (FRACAS / Trouble Reporting).

Focus Area 9: Land-Based Test Site & Shipboard Installation Support

- **Land-Based Test Site (LBTS) Engineering:** Providing on-site systems engineering, software loading, and live test support at locations including the Combat Systems Engineering Development Site (CSEDS) in Moorestown, NJ; and Surface Combat Systems Center (SCSC) in Wallops Island, VA.
- **Major Test Events:** Preparing for and executing critical test events at land-based ranges and on underway ships.
- **Shipboard Installation:** Executing shipboard installation across naval shipyards and bases, including software infrastructure installation, tactical software installation, network configuration, interface checkout, and combat system light-off.

Focus Area 10: Fleet Training Systems & Instructional Design

- Performing Job Duty Task Analysis (JDTA) and Front End Analysis (FEA) to define sailor instructional requirements for modernized tactical systems.
- Developing, maintaining, and updating training curricula, student guides, and tactical job aids.
- Executing and coordinating factory, pilot, and schoolhouse training courses across Surface Combat Systems Training Command (SCSTC) locations.

Focus Area 11: Technical Documentation & Interactive Manuals (IETMs)

- Modernizing traditional technical publication pipelines toward dynamic, S1000D-compliant Interactive Electronic Technical Manuals (IETMs) and Quick Reference Guides (QRGs).
- Providing technical writing support to translate complex software architectures and engineering data into standard operating procedures and maintenance manuals.
- Investigating, adjudicating, and incorporating fleet-submitted Technical Manual Deficiency / Evaluation Reports (TMDERs).

SECTION 5: INSTRUCTIONS TO RESPONDENTS & QUESTIONNAIRE

Interested industry partners are requested to submit responses organized into the following four (4) distinct parts:

Part 1: Corporate Profile & Administrative Information (Max 3 Pages)

1. **Administrative Data:** Company Name, Commercial and Government Entity (CAGE) Code, Unique Entity Identifier (UEI), physical address, website, and designated Points of Contact (POC) for technical and contractual matters.
2. **Business Size & Socioeconomic Status:** Business classification under NAICS 541330; small business designations (e.g., Small Business, 8(a), HUBZone, SDVOSB, WOSB).
3. **Facility & Personnel Clearances:** Current Facility Security Clearance (FSL) level; Secret/Top Secret safeguarding capability; availability of cleared key personnel.
4. **Existing Contract Vehicles:** Active prime contracts (e.g., SeaPort-NxG, GSA OASIS/MAS, or agency-specific MACs) suitable for this scope.

Part 2: Technical Approach, Scope Competency & Innovation (Max 25 Pages)

Respondents must address their specific competencies, technical approaches, and innovative capabilities against the requirements in Section 4. **Vendors may respond to any or all of the functional areas.**

5. **Scope Coverage & Technical Competency:**
    - Clearly delineate which of the eleven (11) Focus Areas your organization has the core capability to execute (as a prime or via teaming).
    - Detail your technical approach, corporate experience, and past performance (within the past 5 years) supporting AEGIS, SSDS, ICS re-architecture, modern weapon and sensor integration, and other real-time / mission-critical engineering and integration efforts.
6. **Super Intelligence (SI) & Advanced Automation Integration:**
    - Describe your organization's specific technical approach to utilizing SI and generative/agentic tools.
    - Provide concrete examples or proposed concepts where SI can be applied across the lifecycle—such as automated test script generation and test execution, automated code refactoring, telemetry and fleet data anomaly detection, MBSE artifact generation, cyber vulnerability triage/STIG scanning, or dynamic IETM/training curriculum updates.
7. **Virtualization Experience:**
    - Detail your experience deploying workloads in virtualized and Kubernetes-orchestrated environments; abstracting hardware dependencies; and

integrating mission critical software in environments demanding high reliability and low latency.

Part 3: Industry Barriers to Entry & Risk Identification (Max 5 Pages)

The Navy is committed to fostering competition, removing friction, and accelerating delivery timelines. Respondents are requested to identify:

8. **Technical & Resource Barriers:** Identify key barriers to entry, technical hurdles, or constraints that may prevent your firm from successfully performing **any** of the 11 focus areas identified above, within required delivery timeframes (e.g., specialized software/hardware tool access, facility/lab access, proprietary interfaces, legacy codebase familiarity).
9. **Government Furnished Information / Property / Computer Programs (GFI/GFE/GFCP):** Specific GFI, GFE, GFCP; developer environment credentials; test facility availability required to de-risk onboarding and ensure parity.
10. **Personnel & Security Bottlenecks:** Security clearance onboarding latency, specialized labor skill shortfalls, or site access restrictions that present schedule risks, along with suggested government mitigation strategies.

Part 4: Acquisition Strategy, Contract Structuring & ROM (Max 7 Pages)

11. **Contract Type & Structuring Recommendations:**

    - Provide industry perspectives and recommendations on optimal contract type (e.g., Cost-Plus-Fixed-Fee, Cost-Plus-Incentive-Fee, Firm-Fixed-Price) and acquisition approach (e.g., Other Transaction Authority (OTA), FAR-based contract, Multiple Award Contract Indefinite Delivery / Indefinite Quantity (MAC IDIQ)). Additionally, address how technical scope could be performed under fixed-price, outcome-based contract line items, along with any perceived risks and tradeoffs of this approach.
    - Discuss how Contract Line Item Numbers (CLINs) or Task Areas should be structured to balance agile, iterative software delivery to Fleet with traditional milestone-driven waterfront installations and systems engineering documentation.
    - Recommend incentive structures (e.g., award fees tied to automated test coverage, escaped defect reduction, or schedule acceleration) that align industry motivation with Navy operational velocity.

12. **Teaming & Subcontracting:**

    - Proposed teaming arrangements, joint ventures, or specialized subcontractors to cover the multi-disciplinary scope.

SECTION 6: SUBMISSION GUIDELINES

13. **Format:**

- Submissions must be delivered in searchable Adobe PDF or Microsoft Word format.
- Standard 8.5 x 11-inch layout, 1-inch margins, minimum 11-point font (graphics, tables, and architectural diagrams may use minimum 9-point font).
- Total submission must not exceed **40 pages** across all four parts (excluding Title Page, Table of Contents, and Acronym Table).

14. **Delivery Instructions:**

- Responses must be submitted via email to the Contracting Officer and Contract Specialist listed below with the subject line:
  "Response to RFI [N6175027R1201] - Surface Combat Systems Engineering and Integration - [Company Name]".
- For attachments exceeding 10MB, respondents must coordinate secure file delivery via the DoD SAFE application (https://safe.apps.mil/).

15. **Submission Deadline:**

- All responses must be received no later than 04 December 2026, 15:00 EST.

SECTION 7: PROPRIETARY MARKINGS & SECURITY HANDLING

Proprietary Markings: If a response contains business-sensitive, trade secret, or proprietary information, it must be explicitly marked on each page. The Government will protect properly marked data from unauthorized disclosure.

Non-Government Evaluators: The Navy may utilize Federally Funded Research and Development Centers (FFRDCs) and System Engineering and Technical Assistance (SETA) contractors to assist in reviewing responses. All support personnel are bound by non-disclosure agreements (NDAs) and organizational conflict of interest (OCI) rules. Respondents wishing to establish proprietary information agreement(s) with non-Government evaluators must clearly declare this in their Part 1 submission.

Responses must be UNCLASSIFIED.

SECTION 8: GOVERNMENT POINTS OF CONTACT

- **Contracting Officer:**
  - Name: Ryan Brophy
  - Email: ryan.p.brophy.civ@us.navy.mil
  - Command: Navy PAE Mission Systems
- **Contract Specialist:**
  - Name: Ashley Mathew
  - Email:ashley.mathew2.civ@us.navy.mil
  - Command: Navy PAE Mission Systems
- **Program Office Representative:**
  - Name: Rick Jandrain
  - Email:richard.r.jandrain.civ@us.navy.mil
  - Command: Navy PAE Mission Systems, PMW 630